

Transportation Disruptions and Welfare: Evidence from a Natural Experiment*

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August 13, 2026

Abstract

We exploit the 2010 eruption of Eyjafjallajökull as a natural experiment to estimate the trade-cost shock generated and the resulting welfare loss for US consumers. Using variation in countries' exposures to the volcanic ash cloud, we estimate the elasticity of imports with respect to flight activity and recover bilateral trade-cost shocks from observed flight cancellations. We then translate these shocks into welfare losses using partial-equilibrium, conditional general-equilibrium, and full general-equilibrium gravity models. The eruption reduced US imports from Europe by 17–24 percent and generated sizeable increases in trade costs. Despite these large trade effects, the implied welfare loss is modest, ranging from 0.11 to 0.48 percent of US absorption.

JEL classification: F14, F17, L93, N74, N75

Keywords: Air cargo, distance data, geographic information systems, gravity model, natural experiment, transportation costs.

*We are grateful to conference participants at the

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1 Introduction

Imagine a temporary disruption to trade. It isn't hard to do. Imagine a naval blockade or the latest natural disaster. When these disruptions occur, what costs do they impose on consumers? Popular characterizations typically build their narratives around qualitative evidence or cite various headline statistics. For instance, the recent closure of the Strait of Hormuz has been framed in terms of its impact on oil prices, commodities markets, and global growth forecasts (The Economist, 2026; OECD, 2026). While such evidence is useful for establishing that disruptions are costly, it provides little insight into the magnitude of the welfare losses involved. Theory offers a framework for bridging that gap. The core idea is that economic impacts of disruptions are derived from a common mechanism that stems from an increase in trade costs with downstream consequences for prices and welfare (Anderson and van Wincoop, 2003; Arkolakis et al., 2012). The key empirical challenge in quantifying welfare losses is therefore credibly estimating the first link in that causal chain: the shock to trade costs. This paper takes up that challenge by exploiting a major disruption to international transportation, providing quantitative evidence on the magnitude of the trade-cost shock and the welfare losses they impose.

The event at the center of our analysis is the eruption of Eyjafjallajökull, a subglacial volcano located in southern Iceland. The eruption imparted a large shock to the European air transportation network in April 2010 through the widespread dispersal of volcanic ash, ultimately resulting in the cancellation of more than 100,000 commercial flights (Eurocontrol, 2010). Although the disruption to air traffic was extensive, it was not accompanied by direct shocks to production. Instead, its immediate effects were confined to the movement of goods (and passengers) through the air network. That is, this event generated a substantial transportation shock while avoiding many of the confounding factors that typically accompany disruptions arising from conflicts or extreme weather events. At the same time, because the disruption to air traffic was unanticipated, concerns regarding anticipatory adjustments to trade behavior are limited. Together, these features make the Eyjafjallajökull eruption a useful case study for studying the economic impacts of transportation disruptions.

And certainly, in this case, those impacts were global. Europe is the world's largest trading bloc. Accordingly, a shock to Europe's air transportation network has the potential to disrupt trade on a global scale. Our analysis, though, focuses only on the transatlantic trade

between Europe and the United States, and consequently only on the welfare losses borne by American consumers. This focus is natural as it reflects two considerations. First, the data on US imports are unusually rich. Detailed information on the mode of transportation is available at various levels of aggregation and at a monthly frequency. Second, Eyjafjallajökull sits below some of the major air corridors linking Europe and North America. As a result, transatlantic routes were particularly exposed to the ash cloud, which provided additional variation in flight activity useful for identifying the trade-cost effects of the disruption.

To motivate the welfare analysis, we use an event-study to document the effects of the eruption on trade flows. This establishes the order of magnitude involved and shows how the effects of the disruption varied with countries' exposure to the volcanic ash cloud. The welfare analysis itself proceeds in three steps. We begin by estimating the elasticity of US air imports with respect to the number of flights. Identifying this relationship is challenging because the volume of trade between countries can be affected by the same factors that make them more closely connected through the air transportation network. For instance, strong cultural ties between countries can generate both more trade and more flights. Additionally, the relationship between flights and trade is inherently simultaneous; increases in flight connectivity can facilitate stronger trade linkages, but it is also feasible that stronger trade can generate additional demand for air services. To overcome this identification problem, we exploit variation in countries' exposure to the volcanic ash cloud. Countries that experienced greater ash exposure also experienced larger disruptions to air traffic, yet the ash cloud itself was unlikely to have affected trade flows except through its impact on transportation. As a result, differences in ash exposure across countries generate plausibly exogenous variation in flight activity that can be used to identify the flight elasticity of trade flows.

Second, we map our estimate of flight elasticity together with a measure of flight cancellations into an estimate of the implied increase in trade costs for each affected country. Intuitively, this step translates the disruption to the transportation network into the key economic object of interest: the trade-cost shock. While it would, in principle, be possible to proceed directly from the reduced-form relationship between ash exposure and trade, the intermediate step through flight disruptions reflects the mechanism through which the eruption affected international trade and ties the estimated trade-cost shock directly to the observed disruption in the air transportation network.

Finally, we translate our estimates of trade-cost shocks into three measures of welfare

loss. We begin with a partial-equilibrium calculation that aggregates bilateral increases in trade costs weighted by trade shares. This yields a direct estimate of the effect on the US price index.

We then move beyond this direct calculation and compute conditional and full general-equilibrium effects (Larch and Yotov, 2018; Anderson et al., 2018). The conditional exercise allows multilateral resistance terms to adjust while holding output and expenditure fixed; the full general-equilibrium exercise additionally allows production levels to respond to the trade-cost shock. Although such adjustments were likely limited given the short-lived nature of the Eyjafjallajökull disruption, they provide useful benchmarks for assessing how transportation shocks may propagate through the wider trading system.

This paper contributes to a small but growing literature on transportation disruptions and their economic consequences. Existing studies have examined a variety of settings and transportation modes. Volpe Martincus and Blyde (2013), for example, study how the 2010 Chilean earthquake disrupted domestic transportation networks and reduced export activity in affected regions. Feyrer (2021) exploits the closure of the Suez Canal between 1967 and 1975 to identify the effects of changes in shipping distances on trade flows. More closely related to our setting, Besedeš et al. (2024) examine the trade consequences of airspace restrictions that force aircraft onto longer and more costly routes. Our paper shares this focus on disruptions to transportation networks but differs in its emphasis on welfare. In this regard, it is related to Hugot and Dajud (2016) and Maurer et al. (2026), who quantify the welfare consequences of hypothetical disruptions to maritime transportation networks within structural gravity models. Our contribution is complementary but distinct. Rather than imposing counterfactual changes in transportation costs, we recover the trade-cost shock from an observed disruption and then translate that empirically identified shock into welfare losses.

More broadly, our paper also connects to the literature linking transportation networks to economic outcomes, for instance, Donaldson (2018) and Baum-Snow (2007). The common theme across these papers is that transportation networks matter because they alter the costs of economic exchange. Our analysis focuses on a temporary disruption rather than a permanent infrastructure investment, but the underlying mechanism linking transportation networks to economic outcomes is broadly similar.

Finally, this paper complements a large literature on the welfare effects of trade barriers.

Indeed, the distinction between transportation disruptions and more traditional trade barriers is, in many respects, secondary. Both operate through the same fundamental channel: an increase in trade costs. We show explicitly that the welfare effects of the Eyjafjallajökull eruption can be expressed in tariff-equivalent terms, facilitating direct comparisons with estimates from the trade-policy literature—a comparison that has become increasingly relevant in recent years. The defining characteristics of transportation disruptions—their abruptness and often short-lived nature—have increasingly come to characterize recent US trade policy as well. This stands in contrast to the historical pattern, in which tariff changes were generally more predictable and reflected relatively persistent shifts in trade policy. Viewed from this perspective, transportation disruptions provide a useful lens through which to study a broader question: what are the welfare consequences of sudden increases in the frictions that separate markets?

The rest of the paper proceeds as follows. Section 2 provides some background on the eruption of Eyjafjallajökull. In section 3, we describe our data. Section 4 presents our estimates of the impact of the disruption on trade, measures the elasticity of trade with respect to the flight activity, and calculates the shock to trade costs. Section 5 conducts our welfare analysis. Finally, section 6 concludes with some remarks.

2 Some Background

Eyjafjallajökull is one of 32 active systems in Iceland that have produced eruptions with varying degrees of explosivity. The 2010 eruption of Eyjafjallajökull fell somewhere near the middle of that distribution. What made Eyjafjallajökull unusual was not its explosivity, but its impact on air transportation. Although disruptions to aviation resulting from volcanic activity are not uncommon per se—for instance, Picquout et al. (2013), document 171 volcanic-ash incidents between 1944 and 2006—typically, disruptions to air traffic are limited and localized. Eyjafjallajökull was different due to the unprecedented scale of its impact.

Several factors combined to produce the largest peacetime disruption to aviation in history. First, the eruption was subglacial—it happened beneath an ice cap. The resulting interaction between magma and ice generated an unusually fine particulate matter (tephra) capable of remaining suspended in the atmosphere for extended periods and being trans-

ported great distances. Second, the location of the volcano mattered. Eyjafjallajökull sits directly beneath the North Atlantic jet stream. This allowed ash ejected from the eruption to be rapidly transported across northern and western Europe.

Finally, the regulatory environment amplified the disruption. Earlier incidents in which aircraft temporarily lost engine power after encountering volcanic ash had encouraged regulators to adopt a near-zero-tolerance standard. Under this framework, forecast ash concentrations above 0.2 mg per cubic meter were sufficient to trigger airspace closures. Together, these conditions set the stage for the disruption that followed.

Eyjafjallajökull began erupting on March 20, 2010. The initial phase of the eruption was effusive; it was characterized by a fissure on the flank of the mountain in an ice-free area. The eruption entered a second and considerably more explosive phase on April 14, when activity shifted to the summit and large quantities of tephra were ejected into the jet stream. Over the following week, the ash plume spread across much of northern and western Europe. The most severe disruptions occurred between April 15 and April 21. By April 22, much of the affected airspace had reopened. Subsequent disruptions during late April and May were comparatively limited in scope and largely confined to the United Kingdom, Ireland, and the Iberian Peninsula.

Approximately 104,000 flights were cancelled during the principal disruption window Eurocontrol (2010). However, the geographic footprint of the disruption was far from uniform. While the United Kingdom, Ireland, and several Nordic countries experienced widespread cancellations, southern Europe was only lightly affected. In total, 112,000 flights continued to operate during the same period, reflecting both temporary breaks in the ash cloud and these marked differences in exposure across countries.

3 Data

Our analysis draws on four sources of data. This includes two datasets on trade flows, one specifically focused on the United States, and another that provides global coverage, which includes both bilateral flows but also domestic sales. In addition, we utilize air-traffic information on flights to the United States, as well as daily maps of the geographic extent of the volcanic ash cloud during April and May 2010. In what follows, we describe each dataset, its principal characteristics, and the construction of our sample, beginning with the

trade data in Section 3.A.

A Trade Data

Our trade data are drawn from two sources. The first is the US Imports of Merchandise dataset, maintained by the US Census Bureau. These data provide detailed information on a single reporter—the US, but on a very detailed level. The second is the International Trade and Production Database for Estimation (ITPD-E), developed by Borchert et al. (2022). These data are specifically designed for structural gravity estimation and general equilibrium trade analysis. Below, we discuss each of these data sources, though both are widely used in the trade literature, so we limit our discussion to features that are relevant for the present study.

A key feature of the US Imports of Merchandise dataset is its monthly frequency. This is particularly important in our setting, as the Eyjafjallajökull disruption was both sudden and short-lived. Second, the data identify the mode of transportation associated with each shipment, allowing us to isolate imports arriving by air. The U.S. Census Bureau reports import values at the 10-digit Harmonized System level. While aspects of our analysis leverage information on these highly disaggregated product-flow data, the bulk of the empirics utilizes the aggregate bilateral import number constructed from the underlying HS10 data.

The sample on which the bulk of our regression analysis is based is composed of a balanced panel of 70 countries. An important feature of this sample is that each country maintained direct air service to the United States throughout the period of study. The sample period starts in January 2008 and extends to July 2012. The eruption, therefore, sits at the midpoint of this window, providing balanced pre- and post-eruption sample windows on either side of the disruption.

While the US Census Bureau reports exports in addition to imports, our analysis focuses exclusively on their import data. This is natural given our objective of quantifying the welfare losses borne by US consumers. Since those losses arise through changes in the prices of imported goods, imports are the relevant margin of analysis.

Exports, however, are essential when trying to recover the general equilibrium welfare effects of trade cost shocks. For this part of the analysis, we rely on the International Trade and Production Database for Estimation. These data provide balanced data on roughly

240 countries, observed at an annual frequency. A crucial feature of these data is that it plugs holes in traditional datasets that make them unsuitable for general equilibrium trade analysis. In particular, the ITPD-E contains information on domestic trade, and fills missing bilateral trade information, and also provides sectoral breakdowns.

B Air Traffic Data

Information on air traffic is obtained from the U.S. Department of Transportation’s T100 database. These data report monthly information on direct flights between the United States and its trading partners, including both actual and scheduled departures and arrivals. The availability of scheduled flight information is particularly valuable for our purposes, as it allows us to compare realized flight activity with the schedule that airlines had planned prior to the eruption, thereby providing a natural measure of the disruption to air transportation.

The T100 database reports information separately for passenger and cargo-only freighters. In principle, the number of cargo flights provides a more direct measure of air cargo capacity. In practice, however, these data are incomplete and contain very limited information on scheduled operations. For this reason, our analysis focuses on passenger flights. This choice is also economically meaningful, as approximately one-half of all air cargo is transported in the belly hold of passenger aircraft. Passenger flight activity therefore, provides a natural proxy for the disruption to air freight capacity generated by the Eyjafjallajökull eruption. As a robustness check, the Appendix redoes parts of our analysis using total flight activity, i.e., the sum of passenger and cargo flights.

C Ash Coverage

Data on the geographic extent of the volcanic ash cloud are obtained from daily advisories issued by the London Volcanic Ash Advisory Centre (VAAC). Our analysis is based on the advisory issued closest to 12:00 UTC each day. The advisories distinguish between three altitude bands: 0–20,000 feet, 20,000–35,000 feet, and 35,000–55,000 feet. The corresponding JPEGs were imported into a Geographic Information System (GIS), from which we constructed a separate measure of ash exposure for each altitude layer.

Our measure of ash exposure is based on the cumulative geographic coverage of the ash cloud within each country. For each day, we calculate the fraction of a country’s land area

covered by volcanic ash. Daily coverage is then summed over the course of the month to produce a cumulative ash-exposure measure. For example, if volcanic ash covered 80 percent of a country’s land area on one day, 70 percent on the next, and 50 percent on the following day, these observations yield 2.0 country-days of ash exposure. This procedure is repeated separately for each altitude band.

Although ash exposure is calculated for all three altitude layers, only the lowest layer (0–20,000 feet) proves to be statistically important for the number of flight departures and arrivals. This is consistent with the operational response of the aviation sector, where low-altitude ash was the principal driver of airport closures and flight cancellations. Finally, although the volcano generated intermittent ash clouds during May 2010, these episodes resulted in only limited disruptions to European air traffic. Accordingly, for much of the analysis, we focus on the ash exposure in April.

4 Trade Impacts

A Lost Trade

Our goal is to quantify the welfare consequences of Eyjafjallajökull’s eruption, but as a first step, it is useful to ask the more direct question: what were the impacts on trade?

Some insights can be had by examining growth statistics for US imports during the eruption window. Some of these statistics are reported in Table 1. From these we can deduce a number of patterns. First, US air imports from Europe dropped sharply, down 24%, relative to the previous month in April 2010—the first month of the disruption. Second, air imports from other regions—East Asia, North and South America, and the rest of the world as a whole—changed comparatively little during this time. Third, there was a soft bounce in May, as imports from Europe recovered, rising by about 9% from their lows in April. Finally, there was no comparable decline in European imports arriving by sea in either the first or the second month of the disruption. This is consistent with the view that the disruption operated through the air transportation network rather than through a broader contraction in trade.

While informative, simple averages inevitably confound the effects of the disruption to air traffic with seasonal and cross-sectional variation in trade data. A natural next step

Table 1: Month-to-Month Growth in U.S. Imports During the Eyjafjallajökull Disruption

Region / Mode	Month-to-Month Growth	
	April	May
Europe (Air)	-0.240** (0.105)	0.090 (0.105)
East Asia (Air)	0.060 (0.116)	-0.040 (0.116)
Americas (Air)	-0.010 (0.088)	0.010 (0.088)
Rest of World (Air)	0.010 (0.097)	-0.020 (0.097)
Europe (Sea)	0.030 (0.104)	0.040 (0.104)

Notes: Entries report the average month-to-month growth in US imports during April and May 2010. Unless otherwise indicated, all figures refer to imports arriving by air. The final row reports imports from Europe transported by sea and serves as a placebo comparison. The numbers in parentheses are the standard deviations of the month-to-month growth series for each sample; by construction this is the same for April and May. *, **, and *** denote statistical significance at the 10, 5, and 1 percent levels, respectively.

is to remove this background variation from the data. Below we fit a gravity model that provides more focused estimates. The model is organized around variations of the following econometric specification:

$$M_{i,t} = \exp [\beta D_{i,t} + \mu_{j,y} + \lambda_t + \gamma_m I(\text{Europe})] u_{j,t} \quad (1)$$

Our estimates, reported in Table 2, are based on a 70-country sample, comprised of both European and extra-European trading partners. The sample is restricted to countries that maintained continuous direct air service to the US throughout the pre-eruption window and whose average monthly air imports exceeded 1 million US dollars. We estimate our model multiplicatively using a Poisson Pseudo Maximum Likelihood (PPML) estimator that accommodates zero trade flows and is consistent in the presence of heteroskedasticity (Silva and Tenreyro, 2006).

The dependent variable, $M_{i,t}$, in these specifications, is the value of US imports from partner i at time t . This is regressed against partner-year dummies, $\mu_{j,y}$, that account

for country-specific heterogeneity. We account for time effects, λ_t , globally in our sample, but also allow for systematic differences between European and non-European partners by controlling for Europe-specific monthly dummies using the interaction $\gamma_m I(\text{Europe})$. Finally, the variable $D_{i,t}$ is the treatment, i.e., it measures the intensity of the disruption caused by the volcano. However, its definition varies across the different specifications reported in Table 2.

In column (1), $D_{i,t}$, is a four-column matrix constructed by interacting a European-region dummy with time indicators that switch on during the disruption window: from March through June, 2010. In effect, what we are estimating here is an event study that traces impact dynamics in the months immediately before, during, and after the disruption.

Table 2: Impact of Volcanic Eruption on US Imports from Europe

	Event Study (1)		Baseline (2)	Robustness (3)	Sea-Based (4)
March	-0.0053 (0.015)	April (t)	-0.1104*** (0.040)	-0.1112*** (0.039)	0.0206 (0.014)
April	-0.1694** (0.068)	April lag, (t-1)	-0.0533*** (0.018)	-0.0656 (0.043)	-0.0062 (0.015)
May	-0.0831*** (0.025)	April lead, (t+1)		-0.0088 (0.010)	
June	-0.0133 (0.036)	May (t)		0.0175 (0.057)	
number of observations	3850		3780	3710	3780
number of countries	70		70	70	70

Notes: The dependent variable in columns (1)–(3) is the monthly value of US air imports (in levels); in column (4) it is the corresponding value of sea imports. In column (1), the explanatory variables are month indicators for March–June 2010 interacted with an indicator for Europe. In columns (2)–(4), it is the logarithm of country-level ash exposure in April and May. The notation $Month(t)$ denotes the contemporaneous effect, $Month(t - 1)$ the one-month lag, and $Month(t + 1)$ the one-month lead. Standard errors are clustered by country. *, **, and *** denote significance at the 10, 5, and 1 percent levels, respectively.

Once common components in the data are absorbed into fixed effects, we find that the decline in air imports from Europe was about 17% in April. This is about 30% smaller than what was implied by the import growth statistics (Table 1), but is nonetheless of the same order of magnitude. Additionally, our results from Table 2 suggest that the disruption to trade persisted into May—when imports were 8% below their expected levels. This is not

inconsistent with the positive month-to-month growth in air imports reported in Table 1. Although imports rebounded in May relative to the exceptionally weak April outcome, they nevertheless remained below the level that would normally be expected for that month.

These estimates provide a measure of the average response of European air trade to the disruption. But since the treatment is binary, the impact is symmetric across all (European) countries. But the disruption to air traffic was not uniform. Some countries, in northern and western Europe, such as the UK and Ireland, were more exposed to volcanic ash and experienced widespread airport closures, while others, particularly countries in southern Europe escaped the disruption altogether. Column (2) provides a sharper analysis that exploits this heterogeneity in treatment intensity by using our country-specific measure of volcanic ash exposure as a continuous treatment variable.

Initially, we define the treatment matrix, $D_{i,t}$, solely in terms of countries' exposure to ash in the month of *April*. Although Eyjafjallajökull remained active through May, lower ash volumes, more favorable wind patterns, and the adoption of less restrictive regulatory standards meant that by May the impact on air traffic was comparatively modest. Our specification nevertheless allows the economic consequences of the April disruption to persist into May by including both contemporaneous and one-month-lagged values of our ash metric. Since our specification logs the treatment, the coefficients should be interpreted as elasticities. However, the coefficient values, -0.110 in April and the lagged effect, -0.053 , are best understood through their implied consequences for trade.

There are 20 European countries that were affected by volcanic ash in April. For each of these countries, Table ?? converts our estimates into implied trade losses. These losses vary substantially with the intensity of ash exposure. For instance, our model predicts a 21% decline in US air imports from the UK in April. By contrast, countries in southern Europe experienced more modest disruptions. Imports from Italy to the US, for example, fell by 7% in the same month.

Our model predicts smaller impacts in May, but these impacts were economically significant nonetheless. The loss in UK-US air trade, for instance, is estimated to have been 10%. One interpretation is that these losses were the result of the resurgence in volcanic activity in May, rather than a carryover effect from the original shock in April. This interpretation, though, is belied by the data; when we explicitly control for May ash exposure in the statistical model [column (3)], its coefficient is small and statistically insignificant, while the

coefficient on the lagged April ash metric changes very little.

Although our findings align closely with what we know about the disruption to air traffic—both its timing and its geographic distribution—it is conceivable that the decline in US-Europe trade in April 2010 was because of some other coincident shock, one that was concentrated more on the northern European states. If that were the case, however, we would also expect to observe a comparable decline in sea-based trade, unless, incredibly, that shock was also specific to the aviation sector. Yet this is not what we observe. Our descriptive analysis already suggested that maritime trade remained largely unaffected during April (Table 1), and the gravity estimates point to the same conclusion. Re-estimating the model using sea imports rather than air imports (column 4 of Table 2) reveals no statistically meaningful relationship between April ash exposure and maritime trade. The estimated April coefficient is, if anything, slightly positive. This is unlikely, though, to reflect substitution from air to sea-based modes, because any efforts to switch modes of transport from air to sea would not show up in the sea-based trade data until May or June. Rather, the absence of a negative contemporaneous effect on sea imports is consistent with the view that the disruption operated primarily through the air transportation network.

Taken together, these results fit a narrative: The volcanic eruption disrupted air traffic; in turn, this reduction in connectivity led to a sharp contraction in air-based trade. However, the results in this section only draw a line from the eruption to trade, skipping over the role that disruptions to air traffic played in shaping those trade outcomes. Understanding this relationship, however, is central for understanding how the disruption affected trade costs, and ultimately, welfare. We turn to this next.

B Air Traffic and Trade

It is self-evident that a constraint on the volume of goods trade is the availability of freight capacities. For air-based trade, those capacities hinge on the number of flights operating between country pairs. Quantifying this relationship between flights and trade, however, presents a number of identification challenges. Changes in flight capacity, for instance, are often a response to demand shocks that also directly affect the amount of trade that countries do. At the same time, trade can itself increase the demand for air cargo, both through dedicated freighters and the belly hold of passenger aircraft, while also stimulating business

travel and other passenger traffic. As a result, a simple regression of trade on measures of air connectivity is unlikely to recover the underlying structural relationship between those two variables.

Table 3: Flight Elasticity of Trade

		PPML (1)	Control Function (2)	Lags (3)
Contemporaneous	Flights	0.1899** (0.078)	1.0698*** (0.259)	1.0651*** (0.292)
Lags	Flights ($t - 1$)			0.0020 (0.077)
	Flights ($t - 2$)			0.1411 (0.118)
Number of observations		3832	3850	3850
Number of countries		70	70	70
First-Stage Regression				
Ash exposure			-0.0978*** (0.023)	
Number of observations			3832	
Number of countries			70	

Notes: The dependent variable is the value of air-based imports (in levels). The variable *flights* is the log of the number of flights from country i to the US. Standard errors are clustered by partner country and are reported in parentheses below the coefficients. *, **, and *** denote statistical significance at the 10, 5, and 1 percent levels, respectively.

Here we propose a two-step approach to address these failures of exogeneity. In the first step, we estimate:

$$f_{i,t} = \delta X_{i,t} + \mu_{j,y} + \lambda_t + \gamma_m I(\text{Europe}) + v_{i,t} \quad (2)$$

where $f_{i,t}$ is the log of flights, from country i to the US at time t , and $X_{i,t}$ is our measure of April ash exposure. By construction $X_{i,t}$ is non-zero only during April 2010. The fixed-effect structure mirrors that of our baseline gravity specification [equation (1)], controlling for partner-year effects, common time effects, and Europe-specific seasonal patterns. This first-stage regression isolates the variation in flights that can be attributed to the dispersion of volcanic ash. Since the spatial distribution of the ash cloud was governed by prevailing atmospheric conditions, this variation is plausibly exogenous, conditional on fixed effects.

In the second step, we re-estimate the gravity model in equation (1). Here the treatment variable, $D_{i,t}$, is constructed from two components: the log of flights, $f_{i,t}$, as well as the residuals from the first-stage regression, $\hat{v}_{i,t}$. These residuals capture the variation in $f_{i,t}$ not explained by the instrument $X_{i,t}$ (or the included fixed effects) and potentially correlated with unobserved determinants of trade. Their inclusion therefore purges the endogenous variation from the flights variable, allowing the remaining variation—identified by the volcanic disruption—to recover the causal effect of flights on trade. Our estimator therefore borrows elements of the control function approach, but applies it to PPML regression.

We find that the relationship between flights and trade is near one-for-one; a one percent reduction in flight activity leads to a roughly one percent reduction in air trade [Table 3, column (2)]. This magnitude is economically intuitive. When flights are cancelled, air-based goods trade cannot simply be rerouted instantaneously by other modes of transport. In the short run, fewer flights therefore translate almost mechanically into lower trade volumes.

This unitary response of trade to changes in flight activity, however, is not immediately apparent in the data. When no attempt is made to control for the residual variation in flight activity, the model conflates endogenous changes in $f_{i,t}$ with exogenous shocks to the variable induced by the volcanic disruption; the result is a coefficient [0.19, column (1)] that is roughly one-fifth the size of the control function estimate.

One concern with our specification is that it does not allow for a dynamic response of trade to changes in flight activity. In practice flight cancellations could have persistent effects. However, a dynamic model [reported in column (3)] provides no evidence for this. The contemporaneous elasticity remains unchanged (1.06) in a specification that allows for lags in the flight variable [column (3)]. At the same time, the lagged coefficients are individually small and statistically insignificant.

To place our elasticity estimates in context, it is useful to combine them with the observed magnitude of the disruption to air traffic during April 2010. On average, 17.8% of scheduled flights were cancelled during the eruption. Given our estimated flight elasticity, this implies an 18.8% decline in air-based trade. This prediction closely matches our previous estimates of trade losses reported in Table 2.

Trade costs. One way to interpret flight cancellations during the eruption is as an increase in iceberg trade costs, $\tau_{i,t}$. This recasts the disruption in the language of structural gravity

and provides the link between observed trade losses and welfare. Here we approach this in two steps. First, we form a country-specific prediction of the log decline in US air imports during the eruption:

$$d \ln \hat{M}_{i,t} = \hat{\beta} \hat{Z}_{i,t},$$

where t is April 2010, $\hat{\beta}$ is the estimated flight elasticity of trade reported in column (2), and $\hat{Z}$ is the log of the predicted shock to flights due to the eruption.

$\hat{Z}$ is obtained using scheduled flights to predict a counterfactual value for flight departures $\hat{f}_{i,t}$ (see Appendix for details). Since flight schedules are fixed before the eruption, they provide a natural benchmark for the number of flights that would have operated absent the disruption. The gap between this counterfactual and the observed number of departures, $\ln f_{i,t} - \ln \hat{f}_{i,t}$, measures the realized reduction in flight activity. We then regress this against each country's exposure to ash in April. Fitted values from this regression form our estimate of $\hat{Z}$, which isolates the component of flight cancellations in April that can be directly linked to the volcanic ash.

In the second step, we recover the implied change in iceberg trade costs from the predicted decline in trade. Under the CES gravity model, this relationship is given by:

$$d \ln \hat{\tau}_{i,t} = \frac{1}{1 - \sigma} d \ln \hat{M}_{i,t}.$$

where σ is the elasticity of substitution among air-traded source countries. Table 4 reports $d \ln \hat{\tau}_{i,t}$ for each of the 20 European countries affected by volcanic ash in April. We report three separate values of $d \ln \hat{\tau}_{ij,t}$ corresponding to three different values of the elasticity parameter: the lower bracket $\sigma = 3$, our preferred pooled estimate $\bar{\sigma} = 4.73$ (the trade-weighted Soderbery (2015) estimate documented below), and the upper bracket $\sigma = 6$.

The trade-cost increases reported in Table 4 are economically meaningful. Under our preferred elasticity of substitution, the disruption raised iceberg trade costs by more than 8% (based on $\sigma = 4.73$) for countries most affected by volcanic ash—the UK and Ireland. At the same time, the shock was highly heterogeneous, as countries further away from the disruption and located further south, such as Italy, experienced a trade cost shock of less than 1%. Below we quantify what these costs imply for the welfare loss borne by US consumers.

Table 4: Estimated Trade Cost Increases by Country (%)

Country	$\sigma = 4.73$	$\sigma = 3$	$\sigma = 6$
GB	8.59	16.03	6.41
IE	8.37	15.61	6.24
DK	7.90	14.72	5.89
BE	7.84	14.63	5.85
NL	7.71	14.39	5.75
DE	6.89	12.84	5.14
LU	6.36	11.87	4.75
CH	5.75	10.73	4.29
CZ	5.66	10.55	4.22
PL	5.60	10.45	4.18
FR	4.93	9.20	3.68
IS	4.48	8.35	3.34
AT	4.29	8.01	3.20
HU	4.08	7.60	3.04
FI	3.67	6.84	2.74
SE	3.44	6.42	2.57
NO	2.68	5.00	2.00
UA	1.74	3.25	1.30
IT	0.69	1.28	0.51
RU	0.12	0.23	0.09

Notes: Entries report the implied proportional increase in iceberg trade costs attributable to the volcanic disruption. Estimates are shown for three alternative values of the elasticity of substitution. Our preferred specification uses $\sigma = 4.73$, the Soderbery (2015) trade-weighted estimate for US air imports from the 20 affected countries observed in 2009.

Calibrating σ . Our preferred value of σ is derived from product-level estimates reported in Soderbery (2015).¹ Specifically, we match each HS-10 product category, k , in our sample to the corresponding elasticity estimate, $\hat{\sigma}_k$, in Soderbery (2015), and compute a trade-weighted average:

$$\sigma = \sum_k w_k \hat{\sigma}_k, \quad (3)$$

where w_k is product k 's share of total air imports in the relevant sample. We compute this weighted average on several samples of data, though our preferred estimate, $\sigma = 4.73$, is calculated on US air imports from affected countries, matching the sample used to identify the trade-cost shock.²

5 Welfare Analysis

We translate the estimated trade-cost shocks into measures of welfare loss. We begin with a partial equilibrium analysis, which captures the direct effect of higher trade costs between the US and affected partners, while holding the broader trading environment fixed. We then extend the analysis to a conditional general equilibrium framework that allows multilateral resistance terms to adjust in response to the disruption. Next, we implement the full general equilibrium framework of Larch and Yotov (2018) and Anderson et al. (2018), which allows both prices and production to adjust endogenously across countries. However, because the disruption lasted only a matter of weeks, production responses are unlikely to play a major role. The full general equilibrium analysis therefore serves primarily as a frame for interpreting our other welfare numbers. Finally, we conclude by reporting the welfare-equivalent tariff—that is, the uniform ad valorem tariff on US imports that would have generated the same welfare loss as the eruption.

¹We use the estimates of Soderbery (2015) rather than those of Broda and Weinstein (2006) because they correct for the small-sample bias and constrained-search inefficiencies present in the earlier methodology. These issues cause the elasticity estimates in Broda and Weinstein (2006) to be overstated.

²For comparison, the corresponding elasticity for sea imports from the same countries is $\sigma^{\text{sea}} = 5.93$. Extending the calculation to all U.S. trading partners yields somewhat different values: $\sigma^{\text{air}} = 4.19$ and $\sigma^{\text{sea}} = 6.880$.

A Partial Equilibrium Analysis

We begin by deriving a first-order approximation to the welfare change under a partial equilibrium closure in which aggregate expenditure is fixed ($d \ln E_{US} = 0$). By implication, welfare changes solely through changes in the domestic price index (the inward multilateral resistance term) P_{US} :

$$\begin{aligned} d \ln W_{US} &= d \ln E_{US} - d \ln P_{US} \\ &= -d \ln P_{US}. \end{aligned}$$

Under the standard structural gravity framework (Anderson and van Wincoop (2003)), P_{US} is:

$$P_{US}^{1-\sigma} = \sum_i \frac{Y_i}{Y} \left(\frac{\tau_i}{\Pi_i} \right)^{1-\sigma},$$

where Y_i denotes total output in country i , Y is world output, τ_i is, as above, the bilateral trade cost between country i and the US, and Π_i is the outward multilateral resistance term.

Totally differentiating this expression and imposing the (partial equilibrium) assumption that outward multilateral resistance is fixed ($d \ln \Pi_i = 0$), yields:

$$d \ln P_{US} = \sum_i \lambda_i d \ln \tau_i,$$

where λ_i is the share of US domestic expenditure devoted to imports from country i . Combining these expressions yields:

$$-d \ln W_{US} = \sum_i \lambda_i d \ln \tau_i, \tag{4}$$

Intuitively, this approach treats the disruption as a direct increase in the cost of imported goods, while holding other prices fixed.³

³An alternative approach for assessing welfare impacts would be to exploit the sufficient-statistics result of Arkolakis et al. (2012) (ACR). Calibrating the ACR statistic, however, requires estimates of the counterfac-

To implement equation 4, we use the estimated country-specific trade-cost shocks, $d \ln \tau_i$ reported in Table 4. The expenditure weights, λ_i , are constructed as the ratio of total imports from country i during the pre-eruption quarter (January–March) to US domestic expenditure over the same period.⁴ Thus, although the disruption affected only the air-traded goods sector, each trade-cost shock is weighted by its importance relative to the US economy as a whole rather than by its importance within the air-traded goods sector.

Table 5 reports the baseline partial equilibrium welfare losses for the three alternative values of σ considered throughout the paper. Using our preferred estimate, $\sigma = 4.733$, the welfare loss to US consumers is approximately 0.11% of domestic expenditure. In 2010, U.S. domestic expenditure on goods was approximately \$8.4 trillion, implying an annualized welfare loss of about \$9.2 billion in 2010 dollars, or roughly \$14.2 billion in today’s terms. Because the eruption disrupted air traffic for only one month, the corresponding one-month loss is approximately \$770 million in 2010 dollars (about \$1.2 billion in today’s purchasing power).

Table 5: Partial Equilibrium Welfare Losses

	$\sigma = 4.73$	$\sigma = 3$	$\sigma = 6$
Welfare change (%)	0.11	0.20	0.08
Annual welfare loss (USD billions)	9.24	16.80	6.72
Monthly welfare loss (USD billions)	0.77	1.40	0.56

Notes: Entries report the partial equilibrium (direct) welfare losses associated with the Eyjafjallajökull eruption. Welfare changes are expressed as percentages of U.S. domestic expenditure. Dollar values are obtained using 2010 U.S. domestic expenditure of \$8.4 trillion; monthly values are one-twelfth of the corresponding annual losses.

One way to think about this is by comparing it to the value of trade displaced by the eruption. Some simple arithmetic proves insightful. In 2010, the United States imported approximately \$1.7 trillion of goods, of which roughly 30% (\$513 billion) arrived by air. About

tual domestic expenditure shares, which are difficult to recover from the estimated trade-cost shocks alone without imposing additional assumptions about how expenditure is reallocated following the disruption. Nevertheless, for completeness, we present an implementation of the Arkolakis et al. (2012) sufficient-statistics approach in the Appendix as a robustness exercise. Reassuringly, despite their different theoretical foundations, the two approaches yield quantitatively similar welfare estimates.

⁴Here we estimate domestic expenditure as the sum of personal consumption *goods* expenditures, gross private domestic investment, and government spending.

11% of this—\$56 billion—originated from Europe. Our estimates suggest that the drop in European imports in April 2010 was about 20%. This implies an annualized trade loss of around \$11.2 billion, or about \$930 million for the one-month loss. The close correspondence between this figure and our estimated welfare loss is interesting, particularly since the calculations rely on entirely different reasoning. The first follows from the structural gravity framework, whereas the second simply applies a dollar value to lost imports.

What is evident is that despite the extraordinary nature of the disruption caused by Eyjafjallajökull, the welfare loss from lost trade was relatively small. But is this altogether surprising? To put our figures into context, it is worth considering estimates of the welfare gains from major trade liberalizations. For example, Caliendo and Parro (2015) find that the long-run welfare gain to the US from NAFTA was approximately 0.08%. The conclusion is striking, but familiar. In fact, a recurring theme in much of the quantitative trade literature is that trade agreements often have large effects on trade, but small welfare effects—especially for large economies. An important caveat is that our estimates abstract from the general equilibrium adjustments that accompany changes in trade costs. We now examine the importance of these adjustments.

B Conditional General Equilibrium

When the eruption raised trade costs between the United States and affected European exporters, it also altered the competitive position of other US partners. As imports from affected countries became more expensive, exporters facing little or no disruption became relatively more attractive suppliers to the US market. Consequently, the welfare impact on US consumers extends beyond the direct effect captured by the partial equilibrium calculation described in equation (4). It also reflects changes in exporters’ access to the US market. To capture these indirect effects, we move to the conditional general equilibrium framework of Larch and Yotov (2018) and Anderson et al. (2018).

In a conditional general equilibrium, aggregate output and expenditure remain fixed, while the multilateral resistance terms are allowed to adjust endogenously. This is a natural approximation for a short-lived disruption such as that caused by the Eyjafjallajökull eruption. Expenditure and incomes are unlikely to move substantively over such a short horizon, but trade can nevertheless be reallocated across suppliers, altering the multilateral

resistance terms that summarize countries' access to international markets.

We implement this counterfactual in steps using the GEPPML framework of Anderson et al. (2018). The method exploits the equivalence between the fixed effects and the multilateral resistance terms (see Fally, 2015). We begin in step one by estimating the following stripped down model:

$$M_{ij} = \exp(\pi_i + \chi_j) \varepsilon_{ij} \quad (5)$$

on the 2009, ITPD-E bilateral trade matrix (Borchert et al., 2022), but restricted to the 146 countries with positive observed internal trade. The notation closely follows that used in the previous section, but is now applied to a full trade matrix (multiple partners *and* reporters); M_{ij} denotes imports from partner i to reporter j , π_i is the partner fixed effect, an estimate of the outward multilateral resistance term, and χ_j denotes the reporter effect, or the estimate of the inward multilateral resistance. Equation (5) delivers a set of baseline exporter and importer fixed effects.

Since the eruption affected only air transport, whereas the ITPD-E reports trade aggregated across all transport modes, we first convert the estimated air trade-cost shock into a trade-cost shock aggregated across transport modes:

$$d \ln \hat{\tau}_{ij}^{\text{agg}} = s_i^{\text{air}} d \ln \hat{\tau}_{ij}^{\text{air}}, \quad (6)$$

where s_i^{air} is the share of bilateral imports from country i transported by air and $d \ln \hat{\tau}_{ij}^{\text{air}}$ is the air trade-cost shock estimated in Section 4.B.⁵

In step two, we re-estimate the gravity model after introducing this bilateral trade-cost shock as an offset in the linear predictor, i.e., we estimate:

$$M_{ij} = \exp[\pi_i^c + \chi_j^c + (1 - \sigma) d \ln \hat{\tau}_{ij}^{\text{agg}}] \varepsilon_{ij}, \quad (7)$$

⁵Our counterfactual perturbs only the bilateral trade costs between affected European exporters and the United States. This reflects the scope of the trade-cost estimates obtained in Section 4.B, which are identified using US import data. In reality, the eruption also increased trade costs between affected European exporters and other destinations. Extending the counterfactual to the full matrix would therefore require an additional assumption that the US-estimated trade-cost shocks are representative of those experienced on other routes. We leave this extension for future work.

where $d \ln \hat{\tau}_{ij}^{\text{agg}} = 0$ for all unaffected country pairs. Re-estimating the equation with this offset produces a new set of exporter and importer fixed effects corresponding to the counterfactual equilibrium. Because the conditional general equilibrium holds output and expenditure fixed, changes in these fixed effects isolate the adjustments in multilateral resistance induced by the eruption.

The conditional welfare change is then given by:

$$d \ln W_j = -d \ln P_j = \frac{\hat{\chi}_j^c - \hat{\chi}_j}{1 - \bar{\sigma}}, \quad (8)$$

where $\hat{\chi}_j$ and $\hat{\chi}_j^c$ denote the importer fixed effects from the baseline and conditional counterfactual estimations, respectively.

Table 6 reports the conditional general equilibrium welfare loss for the United States across the three trade elasticities considered throughout the paper. At our preferred estimate, $\bar{\sigma} = 4.73$, the implied welfare loss is -0.48 percent of domestic expenditure, i.e., an annualized welfare loss of approximately \$40 billion in 2010 dollars, or about \$3.3 billion for the month of April. At the lower elasticity, $\bar{\sigma} = 3$, the welfare loss rises to about -0.9 percent, or roughly \$6.2 billion for the month.

Table 6: Conditional General Equilibrium Welfare Effects

	$\sigma = 4.73$	$\sigma = 3$	$\sigma = 6$
Welfare change (%)	-0.476	-0.888	-0.355
Annual welfare loss (2010 USD billions)	-39.9	-74.7	-29.8
Monthly welfare loss (2010 USD billions)	-3.33	-6.22	-2.49

Notes: Entries report conditional general equilibrium welfare changes calculated using the GEPPML framework of Anderson et al. (2018). The analysis is based on the 2009 ITPD-E bilateral trade matrix (Borchert et al., 2022) and is implemented using the `ge.gravity` procedure of Baier et al. (2019). Welfare changes are recovered from the change in the importer fixed effects according to

$$d \ln W_j = \frac{\hat{\chi}_j^c - \hat{\chi}_j}{1 - \sigma}.$$

Dollar values are calculated using 2010 US domestic expenditure of \$8.4 trillion. Monthly losses are one-twelfth of the corresponding annual values.

The conditional general equilibrium estimate is substantially larger than the corresponding partial equilibrium loss of -0.120 percent. The difference reflects two important features

of the analysis. First, unlike the partial equilibrium calculation, the conditional counterfactual recognizes that a disruption affecting one set of exporters changes the relative competitiveness of every country supplying the US market. The welfare effect therefore reflects not only the direct increase in bilateral trade costs, but also the induced changes in countries' access to international markets summarized by the multilateral resistance terms. Second, because aggregate output and expenditure are held fixed by construction, the entire adjustment must occur through changes in equilibrium prices. As a result, the conditional equilibrium amplifies the welfare effect relative to the direct partial equilibrium calculation.

C Full General Equilibrium

Although the Eyjafjallajökull disruption was short-lived, the estimated trade-cost shocks provide a useful basis for considering a longer-lived event of comparable magnitude. Such counterfactuals are relevant because many contemporary trade disruptions persist for months or even years. In this section, we ask what the long-run welfare implications of a disruption of this magnitude would be in a full general equilibrium model.

Unlike the conditional GE counterfactual—which holds output Y_i and expenditure E_j fixed and lets the multilateral resistance terms absorb the entire shock—the full GE exercise allows Y_i and E_j to adjust endogenously alongside P_j and Π_i . A detailed discussion of the underlying methodology is provided by Anderson et al. (2018) and Larch and Yotov (2018). We implement the full general equilibrium counterfactual using the `ge_gravity` command of Baier et al. (2019), applied to the 2009 International Trade and Production Database for Estimation (ITPD-E) (Borchert et al., 2022).

The bilateral trade cost shock, $d \ln \hat{\tau}_{ij}^{\text{agg}}$ for affected (European) exporters i is calculated as in Section 4.B, i.e., it is the air-based trade cost shock weighted by the air share of country i 's imports to the United States in 2009. Following Baier et al. (2019), this trade cost shock enters the GEPPML algorithm through the bilateral partial-effect coefficient:

$$\beta_{ij} = -(\bar{\sigma} - 1) d \ln \hat{\tau}_{ij}^{\text{agg}}$$

which represents the proportional change in bilateral trade implied by the increase in trade costs. The algorithm then solves for the counterfactual equilibrium values of prices, output, and expenditure consistent with these bilateral trade-cost changes.

Table 7: Full General Equilibrium Welfare Effects for the United States

	$\sigma = 4.733$	$\sigma = 3$	$\sigma = 6$
Welfare change (%)	−0.120	−0.234	−0.088
Annual welfare loss (USD billions)	−10.13	−19.75	−7.43
Monthly welfare loss (USD billions)	−0.84	−1.65	−0.62

Notes: Entries report full general equilibrium welfare changes calculated using the the GEPPML framework of Anderson et al. (2018). The analysis is based on the 2009 ITPD-E bilateral trade matrix (Borchert et al., 2022) and is implemented using the `ge_gravity` procedure of Baier et al. (2019). Dollar values are computed using annual U.S. domestic goods expenditure of \$8.4 trillion in 2010. Monthly values are obtained by dividing the corresponding annual welfare loss by twelve.

Table 7 reports the US welfare change under the full general equilibrium counterfactual for each of the three values of σ . For our preferred elasticity, $\bar{\sigma} = 4.733$, the welfare loss is 0.12 percent of domestic expenditure, corresponding to an annual loss of approximately \$10.1 billion, or about \$840 million each month. Remarkably, these figures are almost identical to the corresponding partial equilibrium estimates reported in Table 5. Thus, allowing output (Y_i) and expenditure (E_j) to adjust endogenously leaves the overall welfare implications of the disruption largely unchanged. While the conditional general equilibrium counterfactual amplifies the welfare loss because it requires the multilateral resistance terms to absorb the entire adjustment. Once output and expenditure are also permitted to respond, these additional adjustment margins generate offsetting reallocations across countries and sectors, bringing the welfare loss back into line with the direct trade-cost channel.

An additional advantage of the full general equilibrium framework is that it identifies the distribution of welfare effects across countries. Table 8 reports the ten largest losers and ten largest gainers. The pattern of welfare losses is intuitive. Ireland experiences by far the largest decline (−1.50 percent), which is unsurprising given its substantial exposure to the ash cloud and the resulting disruption to transatlantic air freight. Switzerland and the United States record the next largest losses, followed by Iceland. Although the UK experienced one of the largest disruptions to air traffic during the eruption, its welfare loss is comparatively modest (8th on our list); thus the magnitude of the welfare effect depends not only on the size of the trade-cost shock but also on the broader structure of production

and trade embodied in the general equilibrium model.

The welfare gains are less straightforward to characterize. While there is no obvious common characteristic shared across the top gainers, the gains appear to accrue primarily to relatively small economies that are largely insulated from the direct trade cost shock. These gains arise through the system-wide reallocation of trade captured by the full general equilibrium model.

Table 8: GE PPML: Top Welfare Losers and Winners ($\bar{\sigma} = 4.733$)

<i>Top 10 welfare losers</i>		<i>Top 10 welfare gainers</i>	
Country	$d \ln W$ (%)	Country	$d \ln W$ (%)
Ireland	−1.499	Republic of Congo	+0.251
Switzerland	−0.139	Chad	+0.218
United States	−0.120	Trinidad and Tobago	+0.175
Iceland	−0.104	Israel	+0.117
Netherlands	−0.084	Venezuela	+0.113
Jamaica	−0.078	Iraq	+0.099
Denmark	−0.066	Nigeria	+0.099
United Kingdom	−0.064	Saudi Arabia	+0.068
St. Vincent & Grenadines	−0.060	Cape Verde	+0.065
Dominican Republic	−0.054	Seychelles	+0.058

Notes: Country-level welfare changes from the full general equilibrium analysis at the preferred trade elasticity, $\bar{\sigma} = 4.733$. The table reports the ten largest welfare losses and gains.

6 Conclusion

This paper exploits a natural experiment to recover the trade-cost shock generated by a major disruption to transportation and then translates that shock into welfare losses within a structural gravity framework. In doing so, it connects two literatures that have largely evolved separately: the empirical literature measuring the trade effects of transportation disruptions and the quantitative trade literature measuring the welfare consequences of changes in trade costs.

Our estimates indicate that despite these sizeable effects on trade and trade costs, the implied welfare losses for the United States are comparatively modest. These results reinforce

a broader finding in the quantitative trade literature: increases in trade costs, at the margin—whether arising from tariffs, transportation disruptions, or other trade barriers—can have substantial effects on bilateral trade flows while producing modest welfare effects, particularly for large, diversified economies.

Our paper is among the first to quantify the welfare consequences of disruptions to international transportation networks. Related contributions include Hugot and Dajud (2016) and Maurer et al. (2026), who examine the welfare implications of counterfactual changes to maritime transportation routes and major global shipping corridors. Our contribution is complementary but distinct. Rather than imposing a counterfactual change in transportation costs, we recover the trade-cost shock directly from an observed disruption and then embed that empirically identified shock within a structural gravity framework to quantify its welfare consequences.

More broadly, this paper provides a framework for assessing the welfare consequences of sharp, but potentially temporary, increases in trade costs. Such shocks have become increasingly salient, not only because of disruptions to transportation networks, but also because of growing volatility in trade policy. Although the sources of these shocks differ, their economic consequences operate through the same fundamental mechanism: an increase in the costs of trading across markets. By recovering the underlying trade-cost shock and expressing its consequences in welfare terms, the framework developed here provides a common basis for evaluating a broad class of future disruptions.

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